SANJANA DISSANAYAKA

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SUMMARY

Computer Science undergraduate with experience in full-stack development and a strong interest in AI and machine learning. Skilled in building modern web applications using JavaScript and React, with backend development in Node.js/Express, .NET/C#, and Spring Boot.(Java).

Experienced with SQL and NoSQL databases including PostgreSQL and MongoDB, and familiar with AI/ML workflows using Python, TensorFlow, and scikit-learn, as well as LLM integration.

Proficient with Git/GitHub, Docker, and Linux environments. Actively engaged in university societies and open-source communities, with a focus on collaboration and continuous learning.

CORE COMPETENCIES

- React
- JavaScript
- TailwindCSS
- Node.js, Express
- .NET (C#)
- Springboot (JAVA)
- SQL, PostgreSQL
- MongoDB, Supabase
- Python
- HuggingFace, Groq API, Google Colab
- Machine Learning, LLM & ML operations
- Git, Docker, CI/CD, Linux

PROJECTS

Trowd – Real-Time Transit Intelligence Platform | Project

- Collaboratively developed a full-stack real-time bus tracking and prediction system using FastAPI, Node.js, React (TypeScript), PostgreSQL, and Redis. Designed an ML-powered ETA Fusion Engine that combines LightGBM and XGBoost predictions with real-time telemetry and historical segment data using confidence-weighted blending for more accuracy.
- Implemented crowd-sourced arrival confirmation pipeline with ML-based detection and temporal validation, integrated with a self-improving feedback loop using real-world data, and enabled low-latency real-time updates via Socket.IO and Redis pub/sub enabling low-latency updates for bus location, ETA and occupancy

AI-Powered Portfolio Assistant with RAG | Project

- Developed an AI-powered portfolio assistant to provide an interactive personalized user experience, enabling users to query information about projects, skills, and experience in real time. Implemented a Retrieval-Augmented Generation (RAG) pipeline using Upstash Vector DB, populated from GitHub and data, with automated updates using cron jobs.
- Implemented semantic search using Hugging Face embeddings and integrated LLaMA 3.1 via Groq API for fast inference, combining vector retrieval with Redis-based chat history and smart summarization. Built the system with a Node.js/Express backend (Upstash Redis) and a React (TypeScript) frontend enhanced with GSAP animations.

AcademiTrend – Academic Analytics & Career Prediction Platform | Project

- Contributed to development of a data-driven academic analytics platform with three ML-based prediction modules: course enrollment, pathway trends, and graduate career/salary forecasting using 12+ models (XGBoost, LightGBM, Random Forest, MLP, Prophet, ARIMA).
- Developed feature engineering pipelines and 20+ RESTful APIs using Flask, supporting automated model evaluation, hyperparameter tuning, trend analysis, and salary visualization.